PROPOSITIONAL LOGIC

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Introduction to Logic

Logic is commonly known as the science of reasoning.

Reasons to Study Logic

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  - Studying basic symbolic logic is like learning a new language.
  - Analyze the logic of ordinary sentences, test arguments for validity, and construct proofs.
  - Exercises are like puzzles, making it enjoyable.

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- Logic Helps You Spot Fallacies & Makes You a Better Citizen:

Propositions

A proposition is any declarative statement that is either true or false, but not both.

We will use lowercase letters, such as $p, q, r, \ldots$ to represent propositions.

We will also use the notation

$$p{:}1 + 3 = 7$$

to denote p to be the proposition $1 + 3 = 7$. The truth value of a proposition is true, denoted by T, if it is a true statement and false, denoted by F, if it is a false statement. Statements that are not propositions include questions and commands.

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   4. How are you? **Solution:** Not a proposition.
   5. $x+3=7$. **Solution:** Not a proposition. It can only become a proposition if Variable x is assigned a value.

Compound Propositions

New propositions called compound propositions or propositional functions can be obtained from old ones by using symbolic connectives which we discuss next. The propositions that form a propositional function are called the propositional variables.

Logical Connectives

In propositional logic, there are 5 basic Connectives. These are:

1. Negation
2. Conjunction
3. Disjunction
4. Conditional
5. Biconditional

A truth table displays the relationships between the truth values of propositions.

Negation

Definition: If p is a statement variable, the negation of p is "not p" or "It is not the case that p" and is denoted $\sim p$. It has the opposite truth value from p: if p is true, $\sim p$ is false; if p is false, $\sim p$ is true.

Truth Table for $\sim p$:

p	$\sim p$
T	F
F	T

Example

Find the Negation of the statement. p :Nairobi is in Kenya - T.
Solution: $\sim p$:It is not the case that Nairobi is in Kenya - F
OR
Nairobi is not in Kenya.

Conjunction

Definition: If p and q are statement variables, the conjunction of p and q is "p and q," denoted $p \wedge q$. It is true when, and only when, both p and q are true. If either p or q is false, or if both are false, $p \wedge q$ is false.

Truth Table for $p \wedge q$:

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Disjunction/Inclusive OR

Definition: If p and q are statement variables, the disjunction of p and q is "p or q," denoted $p \vee q$. It is true when either p is true, or q is true, or both p and q are true; it is false only when both p and q are false.

Truth Table for $p \vee q$:

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Interpretations of Disjunction

By Inclusive OR we mean that students who have taken both calculus and computer science can take the class, as well as those who have taken only one of the two subjects.
OTOH, we use Exclusive OR when we say students who have taken either calculus or computer science but not both can take the class. That is, only those who have taken exactly one of the two courses can take the class.

Avoiding Ambiguity

Mathematicians and logicians avoid possible ambiguity about the meaning of the word *or* by understanding it to mean the inclusive "and/or." The symbol $\vee$ comes from the Latin word *vel*, which means *or* in its inclusive sense.

To express the exclusive *or*, the phrase *p* or *q* but not both is used.

Exclusive Or Symbol and Truth Table

The symbol $\oplus$ was introduced to denote exclusive or, so $p \oplus q \equiv (p \vee q) \wedge \sim (p \wedge q)$.
Truth table for exclusive or:

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

Example 1

Let
P:$5 < 9$
Q:$9 < 7$

Construct the propositions $p \wedge q$, $p \vee q$.

Solution:
$P \wedge Q : 5 < 9 \quad \text{and} \quad 9 < 7$
$P \vee Q : 5 < 9 \quad \text{or} \quad 9 < 7.$

Example 2

Consider the following propositions

$$P : \text{It is Friday}$$
$$Q : \text{It is raining.}$$

Construct the propositions $p \wedge q$, $p \vee q$.

Solution:

It is Friday and it is raining

It is Friday or it is raining.

Order of Operations in Logic

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- In most treatments of logic, the symbols $\wedge$ and $\vee$ are considered coequal in order of operation.
- An expression like $p \wedge q \vee r$ is considered ambiguous and must be written as either $(p \wedge q) \vee r$ or $p \wedge (q \vee r)$ to have meaning.

English Words and Logic Symbols

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- Generally, the word *but* is used in place of *and* when the part of the sentence that follows is, in some way, unexpected.
- Another example involves the words *neither-nor*. If p and q are statements, then;
  p *but* q means $p \wedge q$, and
  neither p *nor* q means $\sim p \wedge \sim q$.

Translating from English to Symbols

Write each of the following sentences symbolically, letting $h =$ "It is hot" and $s =$ "It is sunny".

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1. It is not hot but it is sunny.
   Symbolically: $\sim h \wedge s$.

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Write each of the following sentences symbolically, letting $h =$ "It is hot" and $s =$ "It is sunny".

1. It is not hot but it is sunny.
   Symbolically: $\sim h \wedge s$.
2. It is neither hot nor sunny.
   Symbolically: $\sim h \wedge \sim s$.

And, Or, and Inequalities

Suppose x is a particular real number. Let $p, q,$ and r symbolize:

$$p : 0 < x$$
$$q : x < 3$$
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   Symbolically: $q \vee r$.

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   Symbolically: $q \vee r$.
2. $0 < x < 3$.
   Symbolically: $p \wedge q$.
3. $0 < x \leq 3$.
   Symbolically: $p \wedge (q \vee r)$.

Statement Forms and Truth Tables

A statement form (or propositional form) is an expression made up of statement variables and logical connectives. It becomes a statement when actual statements are substituted for the component statement variables.

Constructing Truth table for compound propositions:

i) Start with the variables in the first column.

ii) If there are n variables, the 2^n rows are required to allow for all possible combinations of T and F.

iii) The truth value of the compound proposition appears in the last column.

Exercise: Constructing Truth Tables

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